# How we measure: radar

16

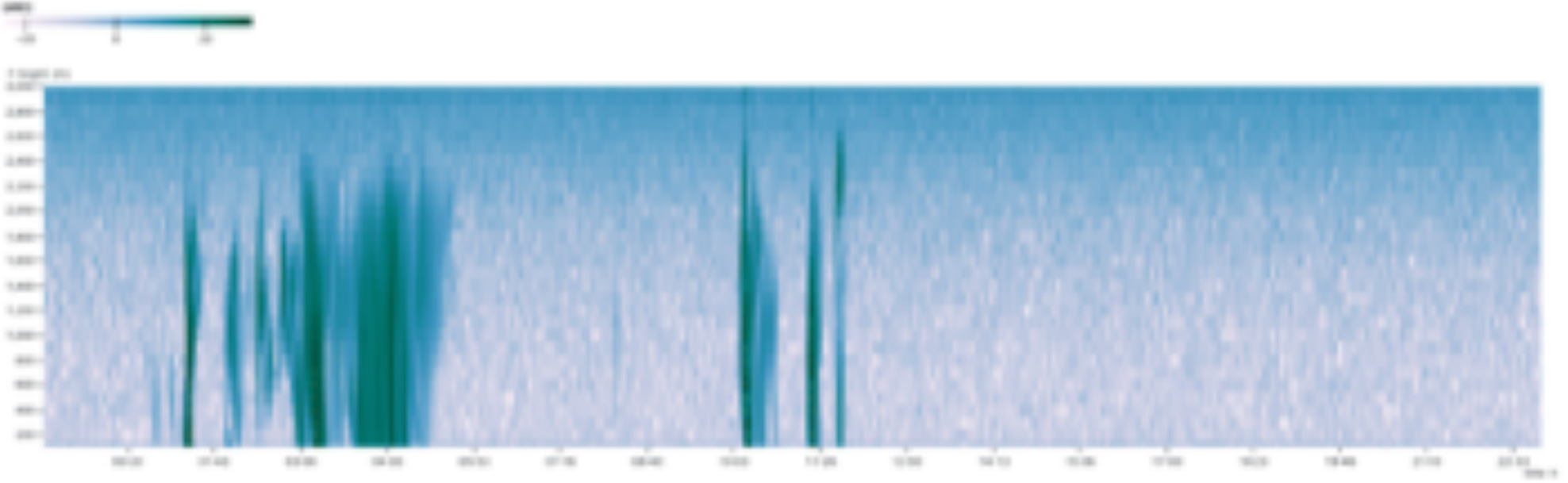
- Radar instruments
  - Similar to LIDAR, but with microwave radiation (larger wavelengths)
  - Sizes and concentrations of water particles

- Use various wavelengths:
    - Ka-band ($\lambda \approx 1\,\mathrm{cm}$), K-band (AKA micro rain, $\lambda \approx 10\,\mathrm{mm}$), and W-band ($\lambda \approx 1\,\mathrm{mm}$) radar

# Our Ka-band radar at the Barbados Cloud Observatory (BCO)

radar reflectivity

[values above (below) range shown in orange (blue)]

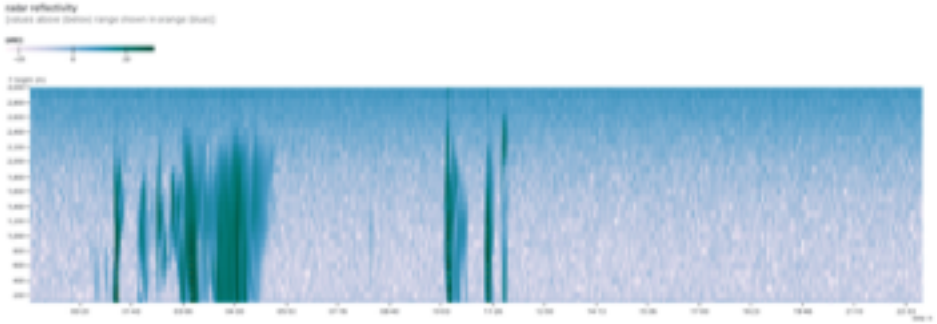


# Radar reflectivity from microwave rain radar (2026-02-08)

# How we measure: radar

- Radar instruments
  - Similar to LIDAR, but with microwave radiation (larger wavelengths)
  - Sizes and concentrations of water particles
- Use various wavelengths:
  - Ka-band ($\lambda \approx 1\,\mathrm{cm}$), K-band (AKA micro rain, $\lambda \approx 10\,\mathrm{mm}$), and W-band ($\lambda \approx 1\,\mathrm{mm}$) radar

Our Ka-band radar at the Barbados Cloud Observatory (BCO)



Radar reflectivity from microwave rain radar (2026-02-08)

# How we measure: rain

A disdrometer at the Barbados Cloud Observatory (BCO)

Rain intensity and drop size distribution plots from three disdrometers at BCO for 2026-02-08